

Gaussian Entanglement Renormalization of the Flattened Domain-Wall Hamiltonian

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We describe a quasi-one-dimensional Gaussian entanglement renormalization procedure for the half-filled ground state of the overcomplete-Wannier flattened Hamiltonian in the domain-wall Chern-insulator geometry. The two-dimensional system is regarded as a one-dimensional chain in the y direction, with each row carrying $q = 2N_x$ complex-fermion orbitals. A binary RG layer groups neighboring rows, diagonalizes the restricted correlation matrix in each two-row block, keeps the natural orbitals with largest $n(1-n)$, and projects the global Gaussian state into the retained subspace. This construction is not a scale-invariant MERA: no disentangler is optimized and the block isometries are allowed to depend on layer and position. Its purpose is instead a smoke test for the MERA hypothesis. If the domain-wall flattened state has the expected infrared structure, local Gaussian mode sorting should polarize gapped bulk modes while retaining mixed modes localized near the two domain walls, and the full- x strip entropy should remain compatible with $c_{\text{eff}} = 1$.

I. FLATTENED HAMILTONIAN TARGET STATE

The microscopic single-particle Hilbert space has dimension

$$N_{\text{layer}} = 2N_x N_y, \quad (1)$$

where the factor of two is the internal Chern-insulator orbital index. We use the same domain-wall benchmark geometry as in the flattened-Hamiltonian analysis: $N_x = 20$, $N_y = 40$, $\alpha_1 = 1$, $\alpha_2 = 30$, $\text{DW} = \text{True}$, and $\text{DW-sector truncation enabled}$. The overcomplete-Wannier spinors are denoted

$$W_{A+}, \quad W_{B+}, \quad W_{A-}, \quad W_{B-}, \quad (2)$$

where A, B label the two trial spinors and $\pm$ labels the Chern-insulator band projector used in constructing the Wannier orbital. After reshaping each family into a matrix whose columns are Wannier orbitals, the flattened single-particle Hamiltonian is

$$H_{\text{flat}} = W_{A+} W_{A+}^\dagger + W_{B+} W_{B+}^\dagger - W_{A-} W_{A-}^\dagger - W_{B-} W_{B-}^\dagger. \quad (3)$$

In the notebook this matrix is explicitly Hermitian symmetrized,

$$H_{\text{flat}} \leftarrow \frac{1}{2} (H_{\text{flat}} + H_{\text{flat}}^\dagger), \quad (4)$$

before diagonalization. Let

$$H_{\text{flat}} u_a = \epsilon_a u_a, \quad \epsilon_1 \leq \epsilon_2 \leq \dots \leq \epsilon_{N_{\text{layer}}}. \quad (5)$$

The half-filled Gaussian ground state is obtained by occupying

$$N_{\text{occ}} = N_{\text{layer}}/2 \quad (6)$$

single-particle modes. Equivalently, its occupied-band projector is

$$P_{\text{occ}} = \sum_{a=1}^{N_{\text{occ}}} u_a u_a^\dagger. \quad (7)$$

If the number of strictly negative eigenvalues is not exactly N_{occ} , the notebook falls back to filling the lowest N_{occ} eigenvectors.

For the RG calculation we use the standard complex-fermion correlation matrix

$$C_{ij} = \langle c_j^\dagger c_i \rangle, \quad C = P_{\text{occ}}. \quad (8)$$

The equivalent ± 1 covariance convention used elsewhere in the repository is

$$G = 2C^* - I. \quad (9)$$

The entanglement calculations below are performed directly with C , because the eigenvalues of a restricted correlation matrix are the occupation numbers entering the Gaussian entropy.

II. ENTROPY FROM A RESTRICTED GAUSSIAN STATE

For any set A of single-particle orbitals, let R_A be the rectangular restriction matrix that selects those orbitals. The restricted correlation matrix is

$$C_A = R_A C R_A^\dagger. \quad (10)$$

If $\lambda_\ell \in [0, 1]$ are the eigenvalues of C_A , the von Neumann entropy of the many-body Gaussian reduced density matrix is

$$S(A) = - \sum_{\ell} [\lambda_\ell \ln \lambda_\ell + (1 - \lambda_\ell) \ln(1 - \lambda_\ell)]. \quad (11)$$

This follows from the Gaussian normal form

$$\rho_A = Z_A^{-1} \exp \left(- \sum_{\ell} \xi_\ell f_\ell^\dagger f_\ell \right), \quad \lambda_\ell = \frac{1}{1 + e^{\xi_\ell}}, \quad (12)$$

where the f_ℓ are natural fermion modes of the subsystem.

The diagnostic subsystem is a full- x strip of height A_y ,

$$A(y_0, A_y) = [0, N_x) \times [y_0, y_0 + A_y) \quad (13)$$

with periodic wrapping in y . In the thermodynamic infrared limit expected for two counter-propagating domain-wall channels, the averaged strip entropy obeys

$$\overline{S}(A_y) = m \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + b, \quad c_{\text{est}} = 3m. \quad (14)$$

For the full- x strip in the physical domain-wall benchmark, the target is

$$c_{\text{est}} \rightarrow c_{\text{eff}} = 1. \quad (15)$$

This value is not the chiral central charge of a single edge. It is the Calabrese-Cardy effective central charge seen by a quasi-one-dimensional cut that intersects both domain-wall channels.

III. SUPERSITE REPRESENTATION

The RG is performed along y . We combine all orbitals in a fixed row into a single supersite. A microscopic orbital index is written as

$$i = (y, \mu), \quad y = 0, \dots, N_y - 1, \quad \mu = 1, \dots, q, \quad q = 2N_x. \quad (16)$$

In the code, μ is the combined (x, σ) index. With this ordering the correlation matrix is a block matrix

$$C_{y\mu, y'\nu}. \quad (17)$$

At RG layer ℓ , the number of supersites is $N_y^{(\ell)}$, while the number of retained orbitals per supersite is kept fixed at $q = 2N_x$. The default sequence is therefore

$$N_y^{(0)} = 40, \quad N_y^{(1)} = 20, \quad N_y^{(2)} = 10. \quad (18)$$

IV. BLOCK NATURAL-ORBITAL RG

A binary RG layer groups neighboring supersites,

$$B_j = \{2j, 2j + 1\}, \quad j = 0, \dots, N_y^{(\ell)}/2 - 1. \quad (19)$$

The corresponding two-row block contains $2q$ fermion modes. Let R_j be the $2q \times qN_y^{(\ell)}$ restriction matrix selecting those modes. The local block correlation matrix is

$$C_{B_j} = R_j C^{(\ell)} R_j^\dagger. \quad (20)$$

It is diagonalized as

$$C_{B_j} = V_j n_j V_j^\dagger, \quad n_j = \text{diag}(n_{j,1}, \dots, n_{j,2q}). \quad (21)$$

The columns of V_j define natural orbitals

$$d_{j,a} = \sum_{\beta=1}^{2q} (V_j)_{\beta a}^\dagger c_{B_j, \beta}, \quad (22)$$

with occupations

$$\langle d_{j,a}^\dagger d_{j,a} \rangle = n_{j,a}. \quad (23)$$

For a Gaussian state, $n_{j,a}$ has a direct entanglement interpretation. If $n_{j,a}$ is close to 0 or 1, then that natural mode is nearly pure inside the block and weakly entangled with the complement. If $n_{j,a} \approx 1/2$, the mode is strongly mixed and carries long-distance entanglement.

The notebook ranks modes using

$$\chi_{j,a} = n_{j,a}(1 - n_{j,a}). \quad (24)$$

This quantity is maximal at $n = 1/2$ and vanishes at $n = 0, 1$. In each two-row block, the q modes with largest $\chi_{j,a}$ are retained. If K_j denotes this set of retained natural-orbital labels, the local isometry is

$$W_j = \begin{pmatrix} | & & | \\ v_{j,a_1} & \cdots & v_{j,a_q} \\ | & & | \end{pmatrix}, \quad a_1, \dots, a_q \in K_j, \quad (25)$$

where $v_{j,a}$ is the a -th column of V_j . Thus

$$W_j^\dagger W_j = I_q. \quad (26)$$

The global isometry is the block direct sum

$$W^{(\ell)} = \bigoplus_j W_j. \quad (27)$$

It maps the fine layer's $qN_y^{(\ell)}$ modes to the coarse layer's $qN_y^{(\ell+1)}$ modes, with

$$N_y^{(\ell+1)} = N_y^{(\ell)}/2. \quad (28)$$

The coarse correlation matrix is

$$C^{(\ell+1)} = \left(W^{(\ell)} \right)^\dagger C^{(\ell)} W^{(\ell)}. \quad (29)$$

This is the central update of the notebook.

It is useful to also keep track of the embedding of each coarse mode into the original microscopic Hilbert space. Define

$$E^{(0)} = I, \quad (30)$$

and recursively

$$E^{(\ell+1)} = E^{(\ell)} W^{(\ell)}. \quad (31)$$

The x -profile plotted in the notebook is

$$p^{(\ell)}(x) = \frac{\sum_{y,\sigma,a} \left| E_{(x,y,\sigma),a}^{(\ell)} \right|^2}{\sum_{x',y,\sigma,a} \left| E_{(x',y,\sigma),a}^{(\ell)} \right|^2}. \quad (32)$$

If the retained subspace is dominated by the domain-wall edge channels, this profile should place enhanced weight near the two domain-wall positions.

V. RELATION TO MERA

The update in Eq. (29) has the same algebraic form as a Gaussian MERA coarse graining after setting all disentanglers to the identity:

$$C^{(\ell+1)} = \left(W^{(\ell)}\right)^\dagger \left(U^{(\ell)} C^{(\ell)} U^{(\ell)\dagger}\right) W^{(\ell)}, \quad U^{(\ell)} = I. \quad (33)$$

A genuine scale-invariant Gaussian MERA would go further. One would optimize local Gaussian unitaries U across block boundaries and local isometries W , then reuse the same tensors at every layer, at least after microscopic transients. In such a construction U and W are variational parameters chosen to minimize the energy, maximize local disentangling, or impose an approximate fixed-point condition on the coarse-grained correlation matrix.

The present RG deliberately avoids that harder variational problem. Its limitations are:

1. no boundary disentangler is optimized;
2. the isometries W_j are determined independently in each block;
3. the isometries are layer dependent;
4. the procedure is diagnostic rather than a compact ansatz for the whole many-body state.

For this reason the notebook should be interpreted as a Gaussian entanglement RG, not as a completed MERA construction.

VI. SCALE-INVARIANT GAUSSIAN MERA PROTOTYPE

The companion notebook `scale_invariant.gaussian.mera.ipynb` implements the first variational step beyond the smoke-test RG. The goal is not to optimize an unconstrained $U(80) \times \text{Stiefel}(80,40)$ ansatz, but to test whether a single repeated Gaussian layer can behave as an approximately scale-invariant coarse grainer for the domain-wall state.

A. Repeated layer geometry

At a fixed RG layer, write the quasi-one-dimensional state as C on N_y supersites with internal dimension $q = 2N_x$. A binary MERA layer has two staggered operations. First, a repeated boundary disentangler $u \in U(2q)$ acts on the pairs

$$(1, 2), (3, 4), \dots, (N_y - 1, 0), \quad (34)$$

where the final pair uses periodic wrapping. The corresponding global unitary is

$$U = \bigoplus_{j=0}^{N_y/2-1} u_{(2j+1, 2j+2)}. \quad (35)$$

Second, a repeated isometry $w \in \mathbb{C}^{2q \times q}$ with

$$w^\dagger w = I_q \quad (36)$$

acts on the block pairs

$$(0, 1), (2, 3), \dots, (N_y - 2, N_y - 1). \quad (37)$$

The global isometry is

$$W = \bigoplus_{j=0}^{N_y/2-1} w_{(2j, 2j+1)}. \quad (38)$$

The single-particle correlation matrix is then updated by

$$C' = W^\dagger U C U^\dagger W. \quad (39)$$

Equation (39) is the finite-correlation-matrix form of a Gaussian MERA layer. Its difference from Eq. (29) is not the algebraic form, but the variational restriction: the same u and w are reused at every block and, in the scale-invariant test, at every layer.

B. Initialization of the repeated isometry

The repeated isometry is initialized from the translation-averaged two-row correlation matrix on the block pairing in Eq. (37). Define

$$\bar{C}_B = \frac{2}{N_y} \sum_{j=0}^{N_y/2-1} R_{2j, 2j+1} C R_{2j, 2j+1}^\dagger, \quad (40)$$

where $R_{a,b}$ restricts to the two supersites a, b . Diagonalize

$$\bar{C}_B = V_B n_B V_B^\dagger, \quad (41)$$

rank the $2q$ natural orbitals by

$$\chi_a = n_a (1 - n_a), \quad (42)$$

and form w from the q columns of V_B with largest χ_a . Thus the repeated isometry is the translationally averaged analogue of the block-dependent natural-orbital map used in the smoke-test RG.

C. Fixed-point covariance objective

To test approximate scale invariance, the notebook compares translation-averaged block correlators before and after coarse graining. For the fine state define

$$\bar{C}_{0r} = \frac{1}{N_y} \sum_{y=0}^{N_y-1} R_y C R_{y+r}^\dagger, \quad (43)$$

where R_y restricts to the q orbitals on supersite y . For the coarse state C' , the analogous object is $\bar{C}'_{0r}$. A scale-invariant layer should map coarse separation r to fine separation $2r$, so the loss is

$$\mathcal{L} = \sum_{r=0}^{r_{\max}} \omega_r \frac{\|\bar{C}'_{0r} - \bar{C}_{0,2r}\|_F^2}{\|\bar{C}_{0,2r}\|_F^2 + \epsilon}. \quad (44)$$

The implementation uses $\omega_r = (1+r)^{-1}$. This objective is gauge-sensitive because the retained coarse modes define an internal basis. Consequently the fixed-point loss is interpreted together with the physical diagnostics: the log-chord entropy coefficient, the entanglement spectrum, and the retained x -profile.

D. Active-subspace disentangler

The first optimized disentangler is restricted to a small active subspace of the boundary pair. Let

$$\bar{C}_\partial = \frac{2}{N_y} \sum_{j=0}^{N_y/2-1} R_{2j+1,2j+2} C R_{2j+1,2j+2}^\dagger \quad (45)$$

be the averaged two-row correlation matrix on the disentangler pairing. Its natural orbitals with largest $\chi = n(1-n)$ define an orthonormal matrix

$$B \in \mathbb{C}^{2q \times m}, \quad B^\dagger B = I_m, \quad (46)$$

with $m = 12$ in the default run and $m = 4$ in smoke tests. A small unitary on this subspace is parameterized as

$$u_{\text{act}} = \exp K, \quad K^\dagger = -K. \quad (47)$$

It is embedded into the full two-supersite disentangler by

$$u = I_{2q} + B(u_{\text{act}} - I_m)B^\dagger. \quad (48)$$

The entries of the anti-Hermitian generator K are optimized with `scipy.optimize.minimize` using the fixed-point loss in Eq. (44). This gives a controlled, low-dimensional variational step toward a true Gaussian MERA without attempting a fragile full-matrix optimization.

E. Acceptance diagnostics

The scale-invariant MERA prototype is considered successful only if several diagnostics agree. The fixed-point loss should decrease after optimizing the active disentangler; the repeated-layer entropy should remain close to $c_{\text{eff}} = 1$; the retained subspace should remain localized near the two domain walls in x ; and the half-chain entanglement spectrum near $\lambda = 1/2$ should be preserved under repeated coarse graining. A reduced loss without these physical signatures would be treated as a gauge artifact, not as evidence for an infrared MERA fixed point.

VII. SELF-CONSISTENT MERA V2

The first scale-invariant prototype is useful because it exposes a failure mode. It applies a microscopic natural-orbital isometry repeatedly, but the output of one layer is not guaranteed to be a pure Gaussian fixed-point state in the same internal gauge. If the discarded modes remain entangled with the retained subspace, the raw compressed matrix

$$C'_{\text{raw}} = W^\dagger U C U^\dagger W \quad (49)$$

has eigenvalues spread through $[0, 1]$, rather than being close to a projector. Repeating this mixed compression can create spurious interval entropy unrelated to the edge CFT. This is precisely the pathology that the second notebook version is designed to diagnose.

A. Raw and purified coarse states

The v2 notebook separates the raw compressed matrix from the pure Gaussian teacher state used for repeated-layer diagnostics. After Eq. (49), one constructs

$$C'_{\text{pure}} = \Pi_{1/2}(C'_{\text{raw}}), \quad (50)$$

where $\Pi_{1/2}$ denotes diagonalizing C'_{raw} and occupying the largest half of its eigenmodes. Thus C'_{pure} is the nearest half-filled projector in the eigenbasis of C'_{raw} . The raw matrix is not discarded: the notebook reports the idempotency defect

$$\delta_{\text{idem}}(C'_{\text{raw}}) = \frac{\|C'^2_{\text{raw}} - C'_{\text{raw}}\|_F}{\sqrt{\dim C'_{\text{raw}}}}. \quad (51)$$

If this defect is large, the layer has not yet learned a true MERA disentangler-isometry pair, even if the purified teacher state has stable long-distance observables.

B. Discarded-sector diagnostics

Let $d \in \mathbb{C}^{2q \times q}$ be an orthonormal complement to the retained isometry w , so that (w, d) is unitary on a two-supersite block. For the post-disentangler local block correlation C_B^U , define

$$C_{KK} = w^\dagger C_B^U w, \quad C_{DD} = d^\dagger C_B^U d, \quad C_{KD} = w^\dagger C_B^U d. \quad (52)$$

The discarded-sector occupations are the eigenvalues $n_{d,a}$ of C_{DD} . A successful layer should make these modes nearly pure and weakly coupled to the retained sector:

$$\bar{\chi}_D = \frac{1}{q} \sum_{a=1}^q n_{d,a}(1 - n_{d,a}) \approx 0, \quad \|C_{KD}\|_F^2 \approx 0. \quad (53)$$

These quantities directly test the local MERA requirement that discarded modes be disentangled, rather than merely removed.

C. Self-consistent isometry update

The v2 notebook also avoids blindly reusing the microscopic w . Starting from the averaged natural-orbital isometry, it performs a teacher iteration:

1. apply one raw MERA layer using the current w ;
2. project the coarse matrix to C'_{pure} ;
3. extract a new averaged natural-orbital isometry w_{next} from C'_{pure} ;
4. align w_{next} to the old output gauge by a Procrustes/polar rotation;
5. damp-update w and re-orthonormalize.

The gauge alignment solves

$$R_* = \arg \min_{R^\dagger R = I_q} \|w_{\text{next}} R - w\|_F, \quad (54)$$

implemented by the polar factor of $w_{\text{next}}^\dagger w$. The damped update is

$$w \leftarrow \text{qf}[(1 - \eta)w + \eta w_{\text{next}} R_*], \quad (55)$$

where qf denotes QR re-orthonormalization and η is the damping parameter.

D. Composite v2 objective

The active-subspace disentangler in v2 is optimized against a composite loss whose leading terms enforce local MERA consistency:

$$\mathcal{L}_{\text{v2}} = \lambda_{\text{idem}} \delta_{\text{idem}}^2 + \lambda_{\text{disc}} \bar{\chi}_D + \lambda_{\text{cross}} \overline{\|C_{KD}\|_F^2} + \lambda_{\text{fp}} \mathcal{L}_{\text{corr}}. \quad (56)$$

The default notebook weights are

$$\lambda_{\text{idem}} = 1, \quad \lambda_{\text{disc}} = 2, \quad \lambda_{\text{cross}} = 1, \quad \lambda_{\text{fp}} = 0.05. \quad (57)$$

The small coefficient of $\mathcal{L}_{\text{corr}}$ reflects the fact that the block-correlator fixed-point loss is gauge-sensitive. For a first MERA smoke test, purity and discarded-sector polarization are more direct indicators that the layer is learning a legitimate coarse graining.

The v2 construction remains a controlled prototype. It is not a full unconstrained optimization over $u \in U(2q)$ and $w \in \text{Stiefel}(2q, q)$. Its purpose is to determine whether the visible domain-wall subspace can be made stable under repeated quasi-one-dimensional Gaussian coarse graining before attempting a more expensive variational MERA.

VIII. FINITE-DEPTH GAUSSIAN MERA ATTEMPT

The follow-up notebook `finite_depth_gaussian_mera.ipynb` removes the strongest assumption in the scale-invariant prototypes: it does not force a single microscopic tensor pair to act at every scale. Instead, each layer has its own Gaussian maps

$$C^{(\ell+1)} = W_\ell^\dagger U_\ell C^{(\ell)} U_\ell^\dagger W_\ell, \quad \ell = 0, 1, \dots, \quad (58)$$

with $U_\ell = \oplus_B u_{\ell,B}$ acting on the odd-even block boundaries and $W_\ell = \oplus_B w_{\ell,B}$ acting on the even-odd blocks that are then coarse grained. In the translationally averaged version used in the notebook, all $u_{\ell,B}$ in a given layer are tied to the same active-subspace unitary u_ℓ , and all $w_{\ell,B}$ are tied to the same two-row isometry w_ℓ . The crucial difference from a scale-invariant ansatz is that u_0, w_0 and u_1, w_1 are learned independently.

For a fixed layer, the notebook first applies a candidate boundary disentangler,

$$C_U^{(\ell)} = U_\ell C^{(\ell)} U_\ell^\dagger, \quad (59)$$

and then forms the averaged two-row block covariance

$$\bar{C}_B^{(\ell)} = \frac{2}{N_y^{(\ell)}} \sum_{j=0}^{N_y^{(\ell)}/2-1} R_{2j,2j+1} C_U^{(\ell)} R_{2j,2j+1}^\dagger. \quad (60)$$

The isometry w_ℓ is extracted only after this disentangling step, by diagonalizing $\bar{C}_B^{(\ell)}$ and retaining the q orbitals with largest $\chi_a = n_a(1 - n_a)$. This ordering is closer to the MERA logic: the disentangler should make the discarded sector locally pure before the isometry chooses which modes to keep.

The active u_ℓ search used here is deliberately modest. An active basis is chosen from the most mixed eigenvectors of the averaged boundary covariance, and random anti-Hermitian generators $K = -K^\dagger$ are exponentiated inside that subspace,

$$u_\ell = I_{2q} + B_\ell (e^K - I_m) B_\ell^\dagger. \quad (61)$$

Each accepted trial lowers the local objective

$$\mathcal{L}_{\text{fd}} = \lambda_{\text{idem}} \delta_{\text{idem}}^2 + \lambda_{\text{disc}} \bar{\chi}_D + \lambda_{\text{cross}} \overline{\|C_{KD}\|_F^2} + \lambda_{\text{fp}} \mathcal{L}_{\text{corr}}. \quad (62)$$

The objective is the same physical test as in Eq. (56), but it is now applied separately at each scale rather than to a repeated fixed-point map.

The full $N_x = 20, N_y = 40$ run gives an instructive outcome. The independent finite-depth natural-orbital flow with $u_\ell = I$ remains nearly pure through two layers, with raw idempotency defects of order 10^{-5} , and its fitted central charge stays close to one:

$$c_{\text{est}}^{(0)} \simeq 1.0000, \quad c_{\text{est}}^{(1)} \simeq 1.0538, \quad c_{\text{est}}^{(2)} \simeq 1.0162. \quad (63)$$

This is the first positive MERA-like result: the catastrophic layer-two entropy blowup of the repeated scale-invariant ansatz is not present once the isometry is allowed to change with scale. The retained x -profile remains peaked near the domain walls, and the discarded natural-orbital occupations are strongly polarized.

The active random disentangler does not yet improve the raw finite-depth flow. With the present small active search it slightly worsens the second-layer entropy drift, while a purified-teacher variant gives $c_{\text{est}} \approx 1$ only because the state is explicitly projected back to a half-filled Gaussian projector after each layer. Thus the main lesson is not that the active optimizer is successful, but that the scale-invariance constraint was the dominant failure mode. A practical next step is to use the learned finite-depth $\{w_\ell\}$ as data: gauge-align them across layers, compare their difference, and only then fit a reusable scale-invariant tensor.

A. Attempted scale-invariant fit from finite-depth tensors

The notebook now performs this last diagnostic explicitly. Let w_0 be the isometry learned for the $40 \rightarrow 20$ step and w_1 the isometry learned for the $20 \rightarrow 10$ step. Since the output basis of each isometry is arbitrary, the first comparison removes the right output gauge by solving

$$R_* = \arg \min_{R^\dagger R = I_q} \|w_1 R - w_0\|_F. \quad (64)$$

Equivalently, if $w_1^\dagger w_0 = U \Sigma V^\dagger$, then $R_* = UV^\dagger$. The notebook records

$$\Delta_{\text{out}} = \frac{\|w_1 R_* - w_0\|_F}{\|w_0\|_F}, \quad \Delta_P = \frac{\|w_1 w_1^\dagger - w_0 w_0^\dagger\|_F}{\sqrt{q}}, \quad (65)$$

together with the principal singular values of $w_0^\dagger w_1$.

For the full run the layer tensors are not close:

$$\Delta_{\text{out}} \simeq 0.875, \quad \Delta_P \simeq 0.957 \quad \text{for the } u = I \text{ finite-depth flow.} \quad (66)$$

Even the active and purified variants only reduce these numbers to roughly $\Delta_{\text{out}} \simeq 0.77\text{--}0.78$ and $\Delta_P \simeq 0.85\text{--}0.86$. This is too large to interpret the two finite-depth layers as small gauge deformations of a single scale-invariant isometry.

As a direct stress test, the notebook constructs an averaged candidate

$$w_* = \text{qf} \left[\frac{1}{2} (w_0 + w_1 R_*) \right], \quad (67)$$

and reuses w_* as a repeated scale-invariant tensor. This fails immediately: for the $u = I$ finite-depth tensors the repeated fit gives $c_{\text{est}} \sim 3.8 \times 10^2$ after one layer and has raw idempotency defect of order 10^{-1} . Thus the finite-depth construction is a positive real-space RG, but it is

not yet evidence for an ordinary one-site-periodic scale-invariant MERA. A viable scale-invariant construction would need either a better coarse-site gauge, a nontrivial repeating disentangler optimized together with w , or a larger unit cell in scale.

B. Canonical gauge, period-two test, and reduced wall channels

The finite-depth notebook was extended with three further tests of this obstruction. First, each coarse state is put in a canonical onsite gauge. For a layer- ℓ covariance, define the translation-averaged onsite and nearest-neighbor matrices

$$\overline{C}_0^{(\ell)} = \frac{1}{N_y^{(\ell)}} \sum_y C_{y,y}^{(\ell)}, \quad \overline{C}_1^{(\ell)} = \frac{1}{N_y^{(\ell)}} \sum_y C_{y,y+1}^{(\ell)}. \quad (68)$$

The unitary G_ℓ is chosen to diagonalize $\overline{C}_0^{(\ell)}$, with residual near-degenerate subspaces fixed by the Hermitian part of $\overline{C}_1^{(\ell)}$. The layer isometry is then compared in the gauge-fixed form

$$\tilde{w}_\ell = (G_\ell^\dagger \oplus G_\ell^\dagger) w_\ell G_{\ell+1}. \quad (69)$$

This asks whether the apparent scale dependence was merely a poor choice of coarse orbital basis.

It was not. For the $N_y = 40$ run the canonical-gauge mismatch between the two finite-depth layers is still order one:

$$\Delta_{\text{out}} \simeq 0.878, \quad \Delta_P \simeq 0.957 \quad (u = I). \quad (70)$$

The active and purified variants only reduce this to $\Delta_{\text{out}} \simeq 0.78\text{--}0.80$. Thus canonical onsite gauge fixing does not turn the finite-depth tensors into a one-tensor fixed point.

Second, an optional `GENTRG_DEEP_NY80=1` run performs $80 \rightarrow 40 \rightarrow 20 \rightarrow 10$, with active disentangler optimization disabled so that the deeper finite-depth natural-orbital flow is affordable. The entropy flow remains good: $c_{\text{est}} \simeq 1.113, 1.059, 1.027$ after the three coarse layers, with raw idempotency defects of order 10^{-5} . However, the period-two hypothesis also fails. Comparing layer 0 and layer 2 in canonical gauge gives

$$\Delta_{\text{out}}^{(0,2)} \simeq 0.869, \quad \Delta_P^{(0,2)} \simeq 0.940. \quad (71)$$

The finite-depth flow is therefore not a simple period-two scale-invariant cycle in this row-orbital basis.

Third, the notebook tests a deliberately simple domain-wall-channel projection: keep only the microscopic orbitals with x within one lattice spacing of each domain wall, giving $q_{\text{eff}} = 12$, purify the projected covariance to a half-filled Gaussian projector, and run the same natural-orbital finite-depth RG. This projection is too crude. The initial projected state already has $c_{\text{est}} \simeq 2$,

and under coarse graining it drifts to values between roughly 3 and 5. Although the retained subspace is by construction wall-localized, the simple real-space wall window does not isolate the single low-energy Dirac channel. A useful reduced theory must be built from Hamiltonian or correlation eigenchannels localized at the walls, not from a hard x -space window.

IX. WHY THIS IS A USEFUL SMOKE TEST

The physical question is whether the flattened-Hamiltonian ground state has an infrared decomposition into gapped two-dimensional bulk modes plus gapless domain-wall modes. The smoke-test logic is:

$$\text{gapped local bulk mode} \implies n \approx 0 \text{ or } 1 \implies \chi = n(1-n) \approx 0, \quad (72)$$

$$\text{IR entangled edge mode} \implies n \approx 1/2 \implies \chi \approx 1/4. \quad (73)$$

Thus a successful local Gaussian mode sort should discard polarized modes and retain mixed modes. The retained subspace should then be localized near the domain walls in x , while the strip entropy should continue to fit the periodic log-chord form with c_{est} close to one.

Failure is also informative. If the retained modes are not wall localized, the full- x entropy slope rapidly drifts,

or the discarded modes are not polarized, then an optimized scale-invariant MERA is premature. The likely fixes would be to include explicit disentanglers, enlarge the local block with buffer rows, first project onto an edge-mode subspace, or increase the system size to reduce finite-size mixing between the two domain walls.

X. NOTEBOOK OBSERVABLES

The notebook records four primary diagnostics at each layer:

1. the averaged full- x strip entropy and the fit $c_{\text{est}} = 3m$;
2. the kept and discarded natural-orbital occupation spectra;
3. the retained-subspace x -profile $p^{(\ell)}(x)$;
4. the half-chain entanglement spectrum near $\lambda = 1/2$.

Together these observables test the minimal necessary ingredients for a future scale-invariant Gaussian MERA: local mode polarization, wall localization of the retained modes, and persistence of the expected $c_{\text{eff}} = 1$ infrared entanglement coefficient.